

Total address space

200.100.10.0/24

(256 addresses)

200.100.10.0	200.100.10.1
200.100.10.2	200.100.10.3
200.100.10.4	200.100.10.5
200.100.10.6	200.100.10.7
...	...
200.100.10.252	200.100.10.253
200.100.10.254	200.100.10.255

Before Subnetting

Partial address spaces

200.100.10.0/25

(128 addresses)

200.100.10.0	200.100.10.1
-	-
-	-
200.100.10.126	200.100.10.127

200.100.10.128/25

(128 addresses)

200.100.10.128	200.100.10.129
-	-
-	-
200.100.10.254	200.100.10.255

After Subnetting



IPv4 Subnetting

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Agenda – Part 1

- IP Addressing Problem
- Subnetting Advantages
- Subnetting
 - On a boundary
 - Within a boundary

IP Addressing: Subnets

Recap – Network/Broadcast/Host Addresses

- Example **198.22.45.105/25**
 - Network Address : 198.22.45.0/25
 - Broadcast Address: 198.22.45.127
 - Host Address Range : 198.22.45.1 -> 198.22.45.126 ($2^7-2 = 126$ hosts)

IP Address: 198.22.45.105 Binary: 11000110.00010110.00101101.01101001

Subnet mask: 255.255.255.128 Binary: 11111111.11111111.11111111.10000000

Network Address: 198.22.45.0/25 Binary: 11000110.00010110.00101101.00000000

Broadcast Address: 198.22.45.127 Binary: 11000110.00010110.00101101.01111111

first Host Address : 198.22.45.1 Binary: 11000110.00010110.00101101.00000001

Last Host Address : 198.22.45.126 Binary: 11000110.00010110.00101101.01111110

Recap – Network/Broadcast/Host Addresses

- Example 10.1.50.143/16
 - Network Address : 10.1.0.0/16
 - Broadcast Address: 10.1.255.255
 - Host Address Range: 10.1.0.1 -> 10.1.255.254 ($2^{16}-2 = 65534$ hosts)

IP Address: 10.1.50.143 Binary: 00001010.00000001.00110010.10001111

Subnet mask: 255.255.0.0 Binary: 11111111.11111111.00000000.00000000

Network Address: 10.1.0.0/16 Binary: 00001010.00000001.00000000.00000000

Broadcast Address: 10.1.255.255 Binary: 00001010.00000001.11111111.11111111

first Host Address : 10.1.0.1 Binary: 00001010.00000001.00000000.00000001

Last Host Address : 10.1.255.254 Binary: 00001010.00000001.11111111.11111110

Recap – Network/Broadcast/Host Addresses

- From Last Slide Deck (Week6, talk 1):

- Network Address
- Broadcast Address
- Number of Hosts
- First host address
- Last host address



IP Address: 209.165.200.224 Binary: 11010001.10100101.11001000.11100000

Subnet mask: 255.255.255.252 Binary: 11111111.11111111.11111111.11111100

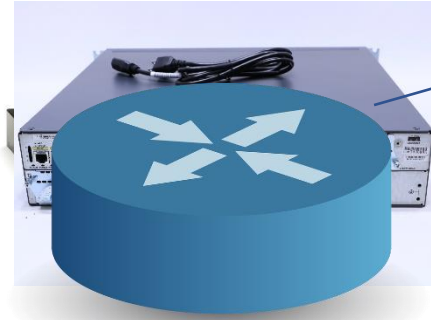
Network Address: 209.165.200.224/30 Binary: 11010001.10100101.11001000.11100000

Broadcast Address: 209.165.200.227 Binary: 11010001.10100101.11001000.11100011

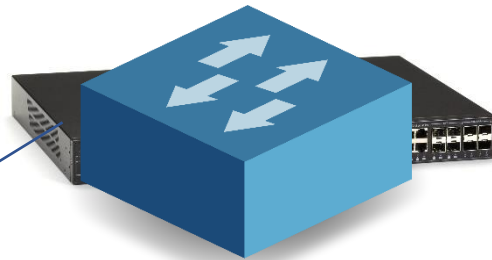
first Host Address : 209.165.200.225 Binary: 11010001.10100101.11001000.11100001

Last Host Address : 209.165.200.226 Binary: 11010001.10100101.11001000.11100010

Number of Hosts: $2^2 - 2 = 2$

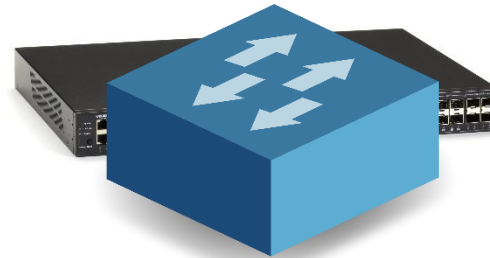


Router

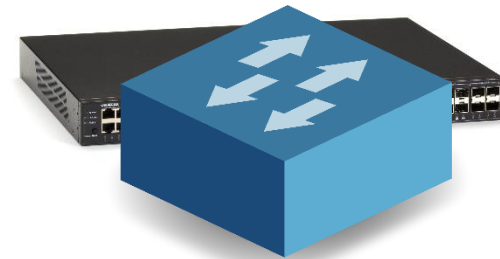


Switch

Shop



Warehouse

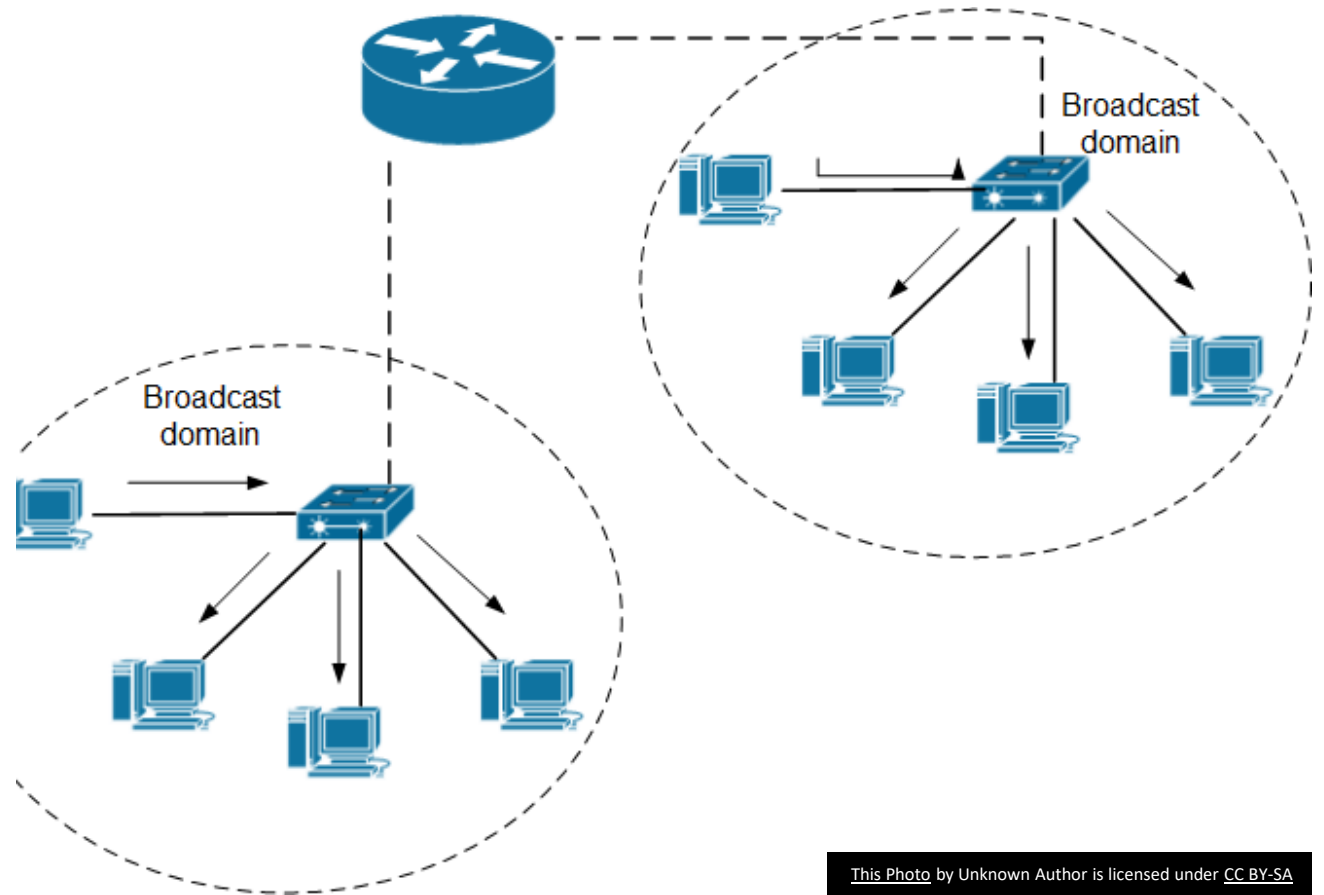


Office



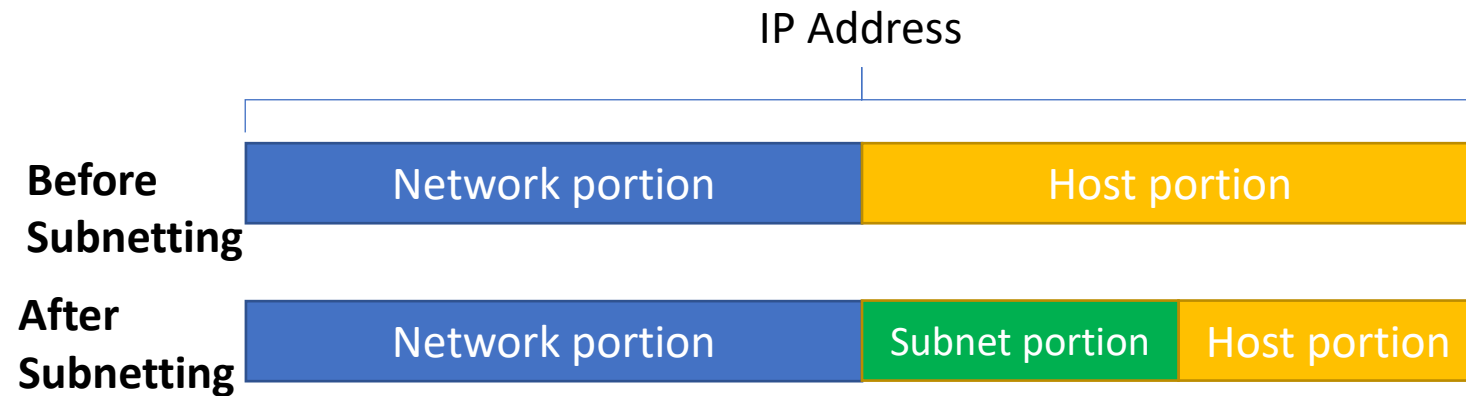
IP Addressing: Problem

- Many protocols (e.g. ARP) need to use broadcast
- As you know – switches send broadcasts on all ports
- **Only device that stops broadcast is a Router**
- **PROBLEM: hosts generating excessive broadcasts and negatively affect the network**



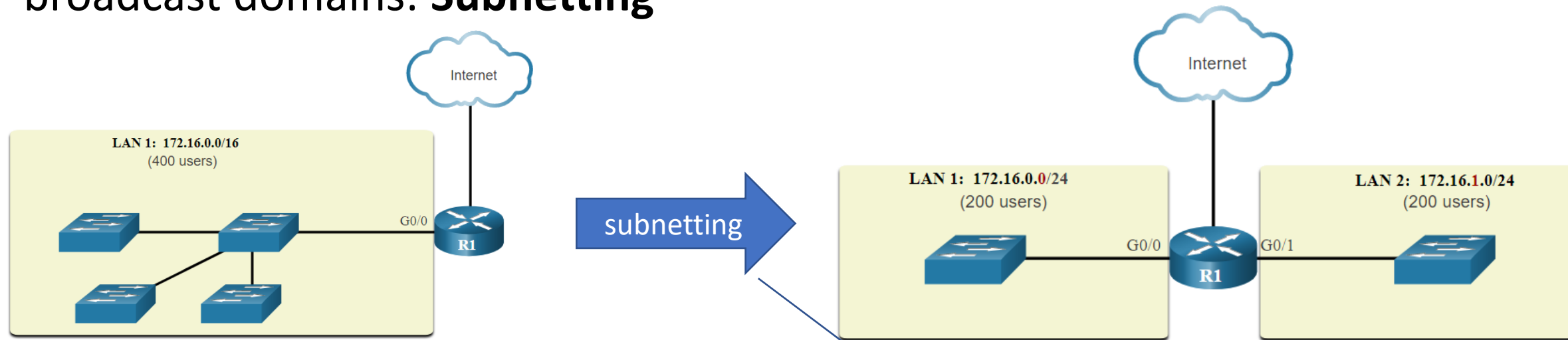
IP Addressing: Broadcast

Reduce the size of the network into smaller **Subnets** to create smaller broadcast domains: **Subnetting**



IP Addressing: Broadcast

Reduce the size of the network into smaller **Subnets** to create smaller broadcast domains: **Subnetting**



Before
Subnetting

Network portion

Host portion

After
Subnetting

Network portion

Subnet portion

Host portion

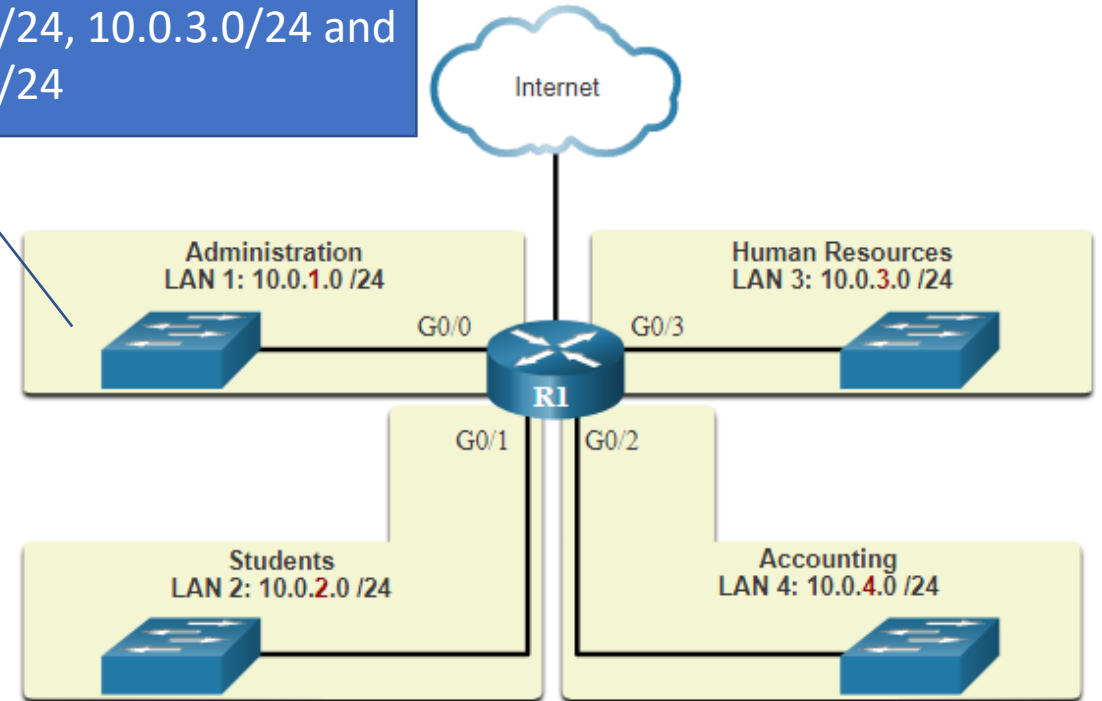
172.16.0.0 /16 is “subnetted” into two subnets of 200 users each: 172.16.0.0 /24 and 172.16.1.0 /24

IP Addressing: Subnetting Advantages

Subnetting advantages:

- Reduces overall network
- Improves network performance
- implement security policies between subnets
- Protects devices from abnormal broadcast traffic

10.0.0.0 /8 is divided into four subnets: 10.0.1.0/24, 10.0.2.0/24, 10.0.3.0/24 and 10.0.4.0/24



IP Addressing: Subnetting on a Boundary

- Networks are most easily subnetted at the **octet boundary** of /8, /16, and /24.
- Example: 10.0.0.0/8 subnetted using /16 and /24

Subnet Address (256 Possible Subnets)	Host Range (65,534 possible hosts per subnet)	Broadcast
10.0.0.0/16	10.0.0.1 - 10.0.255.254	10.0.255.255
10.1.0.0/16	10.1.0.1 - 10.1.255.254	10.1.255.255
10.2.0.0/16	10.2.0.1 - 10.2.255.254	10.2.255.255
10.3.0.0/16	10.3.0.1 - 10.3.255.254	10.3.255.255
10.4.0.0/16	10.4.0.1 - 10.4.255.254	10.4.255.255
10.5.0.0/16	10.5.0.1 - 10.5.255.254	10.5.255.255
10.6.0.0/16	10.6.0.1 - 10.6.255.254	10.6.255.255
10.7.0.0/16	10.7.0.1 - 10.7.255.254	10.7.255.255
...	...	...
10.255.0.0/16	10.255.0.1 - 10.255.255.254	10.255.255.255

Subnet Address (65,536 Possible Subnets)	Host Range (254 possible hosts per subnet)	Broadcast
10.0.0.0/24	10.0.0.1 - 10.0.0.254	10.0.0.255
10.0.1.0/24	10.0.1.1 - 10.0.1.254	10.0.1.255
10.0.2.0/24	10.0.2.1 - 10.0.2.254	10.0.2.255
...	...	...
10.0.255.0/24	10.0.255.1 - 10.0.255.254	10.0.255.255
10.1.0.0/24	10.1.0.1 - 10.1.0.254	10.1.0.255
10.1.1.0/24	10.1.1.1 - 10.1.1.254	10.1.1.255
10.1.2.0/24	10.1.2.1 - 10.1.2.254	10.1.2.255
...	...	...
10.100.0.0/24	10.100.0.1 - 10.100.0.254	10.100.0.255
...	...	...
10.255.255.0/24	10.255.255.1 - 10.255.255.254	10.255.255.255

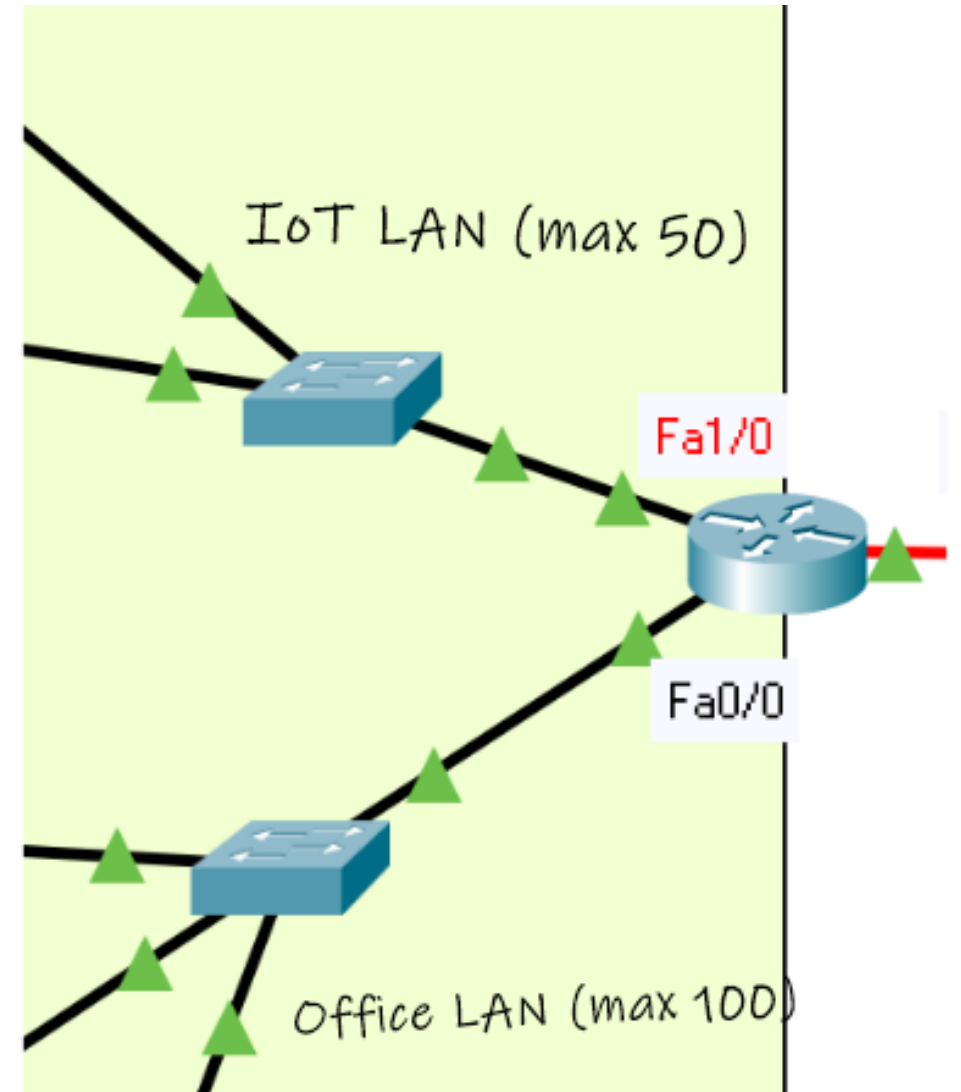
IP Addressing: Subnetting within a Boundary

- Example: subnetting the 192.168.1.0/24 network

Prefix	Subnet mask	Subnet mask (binary, last octet)	# subnets	#hosts
/25	255.255.255.128	.10000000	2	126
/26	255.255.255.192	.11000000	4	62
/27	255.255.255.224	.11100000	8	30
/28	255.255.255.240	.11110000	16	14
/29	255.255.255.248	.11111000	32	6
/30	255.255.255.252	.11111100	64	2

IP Subnetting: Example Problem

- You're asked to design a subnetting solution for a SOHO (small office/home office)
 - Given a network address of 192.168.100.0/24, create a subnet that supports the following:
 - **Create 2 LANs: IoT LAN and Employee LAN**
 - **Hosts on IoT LAN: 50**
 - **Hosts Office LAN: 100**



IP Subnetting: Example Problem

- Select subnet mask that satisfies criteria: **Create 2 LANs: hosts on IoT LAN: 50, hosts Office LAN: 100**

Create Table with all possible subnet masks and choose one that matches criteria.

Prefix	Subnet mask	Subnet mask (binary, last octet)	# subnets	#hosts
/25	255.255.255.128	.10000000	2	126
/26	255.255.255.192	.11000000	4	62
/27	255.255.255.224	.11100000	8	30
/28	255.255.255.240	.11110000	16	14
/29	255.255.255.248	.11111000	32	6
/30	255.255.255.252	.11111100	64	2

- Subnetting 192.168.100.0/24 to /25 gives the following subnets:

Subnet	Host Range	Broadcast
192.168.100.0/25	192.168.100.1->192.168.100.126	192.168.1.127
192.168.100.128/25	192.168.100.129->192.168.100.254	192.168.1.255

Special Use IPv4 addresses

- Network and Broadcast addresses - within each network the first and last addresses cannot be assigned to hosts
- Loopback address - 127.0.0.1 a special address that hosts use to direct traffic to themselves
- Link-Local address - 169.254.0.0 to 169.254.255.255 (169.254.0.0/16) addresses can be automatically assigned to the local host
- There are more – referred to as [Bogon Addresses](#)
 - **Not routed publicly**

Classless Inter-Domain Routing(CIDR)

- Using this approach to Subnetting is also called CIDR
 - Breaks from the old classful addressing approach in the old days
- Used by IANA to issue address “blocks”
- Rules:
 - All IP addresses must be contiguous
 - Block size must be the power of 2 (2^n)
 - First IP address of the Block must be evenly divisible by the size of the block.(always true if the host part of 1st address is all 0s)

Question

- Your Router has the following IP address on FastEthernet0/0:
172.16.2.1/23.
Which of the following are valid host addresses on the LAN connected to FastEthernet0/0.
 1. 172.16.1.99
 2. 172.16.1.198
 3. 172.16.2.255
 4. 172.16.3.0